

Fig. 5 shows the circuit diagram of power supply. The 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 15V, 500mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C1 and then regulated by IC 7812 (IC3). Capacitors C2 and C3 bypass the ripples present in the regulated supply. LED1 acts as a power-on indicator and resistor R7

limits the current through LED1. Diode D5 is used to provide protection to the regulator IC in case its input shorts to ground.

An actual-size, single-side PCB for H-bridge driver using PC is shown in Fig. 7 and its components layout in Fig. 8.

Software

The source program for H-bridge driver using PC is written in Visual

Basic, which is easy to understand. To use the Windows platform, the input32.dll file should reside in the same folder as the executable (.exe) file of the program. The main screen for H-bridge driver using PC is shown in Fig. 9. Effects of screen button along with motor operation are also shown in the table.

EFY note. All the relevant files of this project have been included in this month's EFY-CD.

FORM1.FRM

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VERSION 5.00
Begin VB.Form Form1
    BackColor = &H80000007&
    Caption = "H-bridge Driv-
er"
    ClientHeight = 4515
    ClientLeft = 60
    ClientTop = 450
    ClientWidth = 5730
    Icon = „Form1.frx“:0000
    LinkTopic = „Form1“
    ScaleHeight = 4515
    ScaleWidth = 5730
    StartUpPosition = 3 , Windows De-
fault
    Begin VB.CommandButton Command5
        Caption = „Exit“
        Height = 375
        Left = 2040
        TabIndex = 4
        Top = 3360
        Width = 1455
    End
    Begin VB.CommandButton Command4
        Caption = „Power OFF“
        Height = 375
        Left = 2040
        TabIndex = 3
        Top = 2760
        Width = 1695
    End
    Begin VB.CommandButton Command3
        Caption = „Brake“
        Height = 375
        Left = 2040
        TabIndex = 2
        Top = 2160
        Width = 1695
    End
    Begin VB.CommandButton Command2
        Caption = „Anticlock-
wise“
        Height = 495
        Left = 3240
        TabIndex = 1
        Top = 1200
        Width = 2055
    End
    Begin VB.CommandButton Command1
        Caption = „clockwise“
        Height = 495
        Left = 360
        TabIndex = 0
        Top = 1200
        Width = 2055
    End
    Begin VB.Label Label5
        BackColor = &H00000000&
        Caption = „Brake On/
Off“
        ForeColor = &H00FFFFFF&
        Height = 495
        Left = 3240
        TabIndex = 9
        Top = 480
        Width = 615
    End
    Begin VB.Label Label4
        BackColor = &H00000000&
        Caption = „Power (I)“
        ForeColor = &H00FFFFFF&
        Height = 255
        Left = 2160
        TabIndex = 8
    End
    Top = 480
    Width = 615
    Begin VB.Label Label3
        BackColor = &H80000007&
        Caption = „ACW“
        ForeColor = &H00FFFFFF&
        Height = 255
        Left = 4680
        TabIndex = 7
        Top = 375
    End
    Begin VB.Shape Shape4
        BackColor = &H80000001&
        BorderColor = &H008080FF&
        BorderWidth = 3
        FillColor = &H00FFFFFF&
        FillStyle = 0 , Solid
        Height = 255
        Index = 3
        Left = 3240
        Shape = 3 , Circle
        Top = 120
        Width = 255
    End
    Begin VB.Shape Shape3
        BackColor = &H80000001&
        BorderColor = &H008080FF&
        BorderWidth = 3
        FillColor = &H00FFFFFF&
        FillStyle = 0 , Solid
        Height = 255
        Index = 2
        Left = 2160
        Shape = 3 , Circle
        Top = 120
        Width = 255
    End
    Begin VB.Shape Shape2
        BackColor = &H80000001&
        BorderColor = &H008080FF&
        BorderWidth = 3
        FillColor = &H00FFFFFF&
        FillStyle = 0 , Solid
        Height = 255
        Index = 1
        Left = 4680
        Shape = 3 , Circle
        Top = 360
        Width = 255
    End
    Begin VB.Label Label2
        BackColor = &H80000007&
        Caption = „CW“
        ForeColor = &H8000000E&
        Height = 255
        Left = 600
        TabIndex = 6
        Top = 720
        Width = 375
    End
    Begin VB.Shape Shape1
        BackColor = &H80000001&
        BorderColor = &H008080FF&
        BorderWidth = 3
        FillColor = &H00FFFFFF&
        FillStyle = 0 , Solid
        Height = 255
        Index = 0
        Left = 600
        Shape = 3 , Circle
        Top = 360
        Width = 255
    End
    End
    Attribute VB_Name = „Form1“
    Attribute VB_GlobalNameSpace = False
    Attribute VB_Creatable = False
    Attribute VB_PredeclaredId = True
    Attribute VB_Exposed = False
    Option Explicit
    Dim Value As Integer
    Dim PortAddress As Integer
    Private Sub Command1_Click()
        Value = 1
        Out PortAddress, Value
        Shape1(0).FillColor = vbRed
        Shape2(1).FillColor = vbWhite
        Shape3(2).FillColor = vbRed
        Shape4(3).FillColor = vbWhite
    End Sub
    Private Sub Command2_Click()
        Value = 2
        Out PortAddress, Value
        Shape1(0).FillColor = vbWhite
        Shape2(1).FillColor = vbRed
        Shape3(2).FillColor = vbRed
        Shape4(3).FillColor = vbWhite
    End Sub
    Private Sub Command3_Click()
        Value = 3
        Out PortAddress, Value
        Shape1(0).FillColor = vbWhite
        Shape2(1).FillColor = vbWhite
        Shape3(2).FillColor = vbWhite
        Shape4(3).FillColor = vbWhite
    End Sub
    Private Sub Command4_Click()
        Value = 0
        Out PortAddress, Value
        Shape1(0).FillColor = vbWhite
        Shape2(1).FillColor = vbWhite
        Shape3(2).FillColor = vbWhite
        Shape4(3).FillColor = vbWhite
    End Sub
    Private Sub Command5_Click()
        Value = 0
        Out PortAddress, Value
        Unload Me
    End Sub
    Private Sub Form_Load()
        PortAddress = &H378
    End Sub

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